

Hejun Huang

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SUMMARY

Research Engineer specializing in large-scale recommendation systems and LLM-native platforms. Experienced in building low-latency retrieval and ranking pipelines, streaming deduplication services, and scalable AWS-based systems for high-throughput user-facing and AI native applications.

WORK EXPERIENCE

Amazon.com Services LLC - Software Dev Engineer
Prime Video Personalization & Discovery

Oct 2024 – Present
Seattle, WA

Real-Time Retrieval, Ranking and Recommendation Infrastructure

- Designed, deployed, and operated a distributed recommendation service serving **400M+ users**, handling **80K QPS** across 7 global marketplaces at **99.9% availability** on multi-region ECS/Fargate.
- Ran several A/B experiments across **1M+ concurrent users** and shipped cross-ranker score blending that drove **+1.8% streaming hours** and **+3.5% CTR**.
- Shipped a post-ranking deduplication module with **<5ms p99 overhead**, improving content diversity; validated via shadow traffic and canary rollout.
- Diagnosed and eliminated **15% JVM p99 latency** via memory tuning under high-concurrency traffic.

AI Agent Platform for Production Operations

- Architected a multi-agent incident resolution system (LangGraph + OpenAI SDK) with fan-out decomposition and self-correction loops, cutting resolution time **75%** (3.2d → 8h) at **95%+** accuracy across 100+ escalations.
- Built a RAG knowledge base (Bedrock KB + OpenSearch, hybrid vector + BM25) and Aurora/Redis agent memory layer, achieving **95% retrieval accuracy** at **\$122/mo/team** with 6 teams onboarded in 2 days.
- Built a Root Cause Analyzer that generates Athena SQL over telemetry data and correlates CloudWatch anomalies with deployment history, enabling **sub-10s diagnostics** compared to minutes of manual log triage.

Observability and Data Infrastructure for Production ML Systems

- Built a Spark/Airflow telemetry pipeline processing **30+ TB/day** at **<10s** query latency and **\$0.001/query** with AWS Glue, Athena, and Cradle, achieving a **5,000× cost reduction** over direct Athena scans.
- Instrumented **10+ microservices** with multi-dimensional CloudWatch/Grafana metrics (marketplace × page × content × device), cutting anomaly detection time by **60%**.
- Designed and published reusable CDK monitoring constructs, reducing per-service monitoring code by **75%** and new team onboarding from **2 days to 2 hours**.

Baidu USA - Research Engineer
Robotics and Autonomous Driving Lab

Jan 2024 – July 2024
Sunnyvale, CA

Kubrick: Multi-agent Text-to-Video Generation [paper][demo]

- Built scene graph decomposition and LLM-to-3D rendering pipeline for Kubrick, a multi-agent text-to-video system (GPT-4 + Blender-API) for structured multi-shot generation; published at **CVPR 2025 Workshop**.

EDUCATION

Master in Aerospace Engineering University of Michigan, Ann Arbor

Aug 2022 – Dec 2023

Master in Mechanical Engineering The Chinese University of Hong Kong

Aug 2019 – Dec 2020

Bachelor in Mechatronics Engineering North China Electric Power University

Aug 2015 – Jun 2019

SKILLS

Languages: Java, Kotlin, Python, C++, TypeScript, SQL

Distributed Systems: Kafka, Redis, gRPC, Kubernetes, Docker

AI & LLM: LangGraph, RAG, OpenAI SDK, Vector Databases (Milvus, OpenSearch), JAX, PyTorch

Data & Observability: Apache Spark, Flink, Athena, CloudWatch, Grafana

Cloud & DevOps: AWS, Lambda, SQS, SNS, DynamoDB, MongoDB, MySQL, CI/CD, Bazel

PERSONAL PROJECT

OpenClaw Personal Workflow Agent

01/2024 - Present

- Designed and implemented a **self-hosted multi-agent workflow system** using **OpenClaw**, orchestrating task planning, document analysis, and decision workflows through configurable agents, tool execution pipelines, and structured output generation.
- Integrated **Obsidian-based Markdown notes** as persistent memory, parsing structured task metadata (deadlines, tags, priorities) and generating prioritized daily execution plans delivered as structured Markdown reports.
- Processed structured **health and activity logs** exported from **Apple Health** into CSV format, implementing ingestion and trend analysis pipelines to generate daily activity and recovery recommendations.
- Built domain-specific agents for **resume optimization**, **job description parsing**, and **daily workflow generation**, implementing structured pipelines to extract required skills from job descriptions, match them against resume content, and generate prioritized action plans using **Obsidian Markdown** task data.
- Containerized the system using **Docker** and **docker-compose**, orchestrating agent runtime, data storage, and scheduled execution services to support reproducible deployments and automated daily workflow processing.

PUBLICATION

- [1] He, L. & Song, Y. & **Huang, H.**, et al.. (2025, Nashville, TN) Kubrick: Multimodal Agent Collaborations for Synthetic Video Generation. in *CVPR 2025 AI4CC Workshop*.
- [2] **Huang, H.**, et al.. Privacy-Aware Coverage Design and Analysis in Drone Remote Identification Systems. *IEEE Transactions on Intelligent Transportation Systems*, 2024.
- [3] **Huang, H.**, et al.. (2023, Savannah, GA) Remote Identification Trajectory Coverage in Urban Air Mobility Applications. in *Proceedings of Air Traffic Management R&D Seminar*.
- [4] Li, Z. & **Huang, H.**, et al.. Combining reinforcement learning and barrier functions for adaptive risk management in portfolio optimization. arXiv.
- [4] **Huang, H.**, et al.. (2022, Trieste, Italy) Barrier Certified Safety Learning Control: When Sum-of-Squares Programming Meets Reinforcement Learning. in *Proceedings of Conference on Control Technology and Applications*.
- [5] Han, D. & **Huang, H.**. (2022, Jeju Island, Korea) Sum-of-Squares Program and Safe Learning On Maximizing the Region of Attraction of Partially Unknown Systems. in *Proceedings of Asian Control Conference*.